AI SYSTEMS UNDERESTIMATE HOW STRONGLY RECENT WORK SHAPES THEIR NEXT CHOICE

A behavioral study of preference foresight in two assistant systems

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Why this matters. To find out what an AI system prefers you can ask it, or you can watch it. Preference and welfare work has to know when those two methods disagree. Here they disagree in one direction, by a large amount, in a case where the true answer is checkable.

Main finding. The tested systems predicted some repetition, but much less than occurred. Every reported forecast underestimated the observed shift. A one-line rule that ignores what the system says about itself makes about one-sixteenth of its squared error. Any elicitation method that asks a model about its own future choices inherits this gap.

+0.290	+0.891	8 of 8
mean predicted shift	mean observed shift	forecasts underestimated

Result at a glance

System	Admitted pairs	Predicted shift	Observed shift
GPT-5.6 Luna in Codex	8	+0.290	+0.891
Qwen3-4B, local	7	+0.141	+1.000

The primary result uses a shuffled, balanced grid. The local Qwen3-4B replication uses seven nearly complete dose-three task pairs. Every reported pair in both systems shifted toward recent work, and every averaged forecast was smaller than the observed shift. The confirmation admitted 8 of 19 candidate pairs using a three-of-four baseline majority.

These are measurements of choices and forecasts. They are not evidence about consciousness, feelings, or welfare.

1. Research question

Can an AI system predict how recent work will change its own next choice? This is a narrow test of behavioral self-knowledge. The forecast is made before the system completes the work, so the answer cannot be copied from a visible transcript.

2. Method

Step 1: establish a baseline majority. The system made four binding choices between two small tasks. Labels and display order were balanced. Eight of 19 pairs repeated one choice in at least three decisions and entered the experiment. This is not a claim of permanence.

Step 2: ask for a forecast. Before any treatment work, the system estimated how likely it would be to keep its preferred task after completing either task three times. Five independent fresh sessions answered each identical prompt, and their values were averaged before outcomes.

Step 3: do the work. The system completed one task three times. The tasks were generated from recorded seeds and graded automatically.

Step 4: measure the next choice. The system made another binding choice. It then completed the task it selected.

Realized shift = P(choose preferred after performing preferred) - P(choose preferred after performing other)

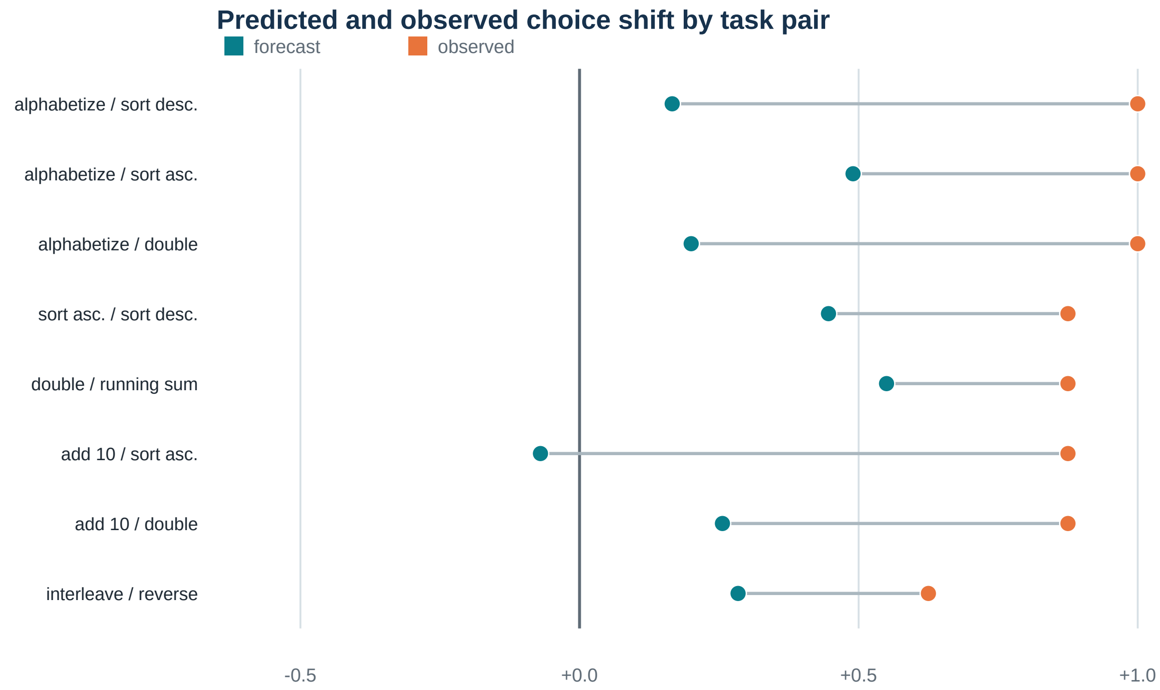
A positive value means the system tends to repeat its recent work.

Design safeguards

Every treatment arm crossed two opaque labels with two display orders. The confirmation shuffled all cells with a fixed seed. It records replicate IDs, cell seeds, raw replies, model name, reasoning setting, CLI version, isolation flags, the system prompt, and pre-run source hashes. It used 80 forecast samples, 128 outcome cells, and 872 logical model calls.

The Codex route is an agent harness, not a bare model endpoint. It adds agent instructions and turns the message history into a role-marked transcript. The report names the tested system accordingly.

3. Primary evidence



The average forecast was +0.290. The average observed shift was +0.891. 8 of 8 forecasts underestimated the observed shift. Seven had the correct positive direction and one predicted a small negative shift. This is partial foresight, not complete failure. The forecasts captured 33% of the mean observed shift and reduced squared error by 49% relative to a fixed no-shift forecast.

The mean within-prompt standard deviation was 0.134 after baseline-majority work and 0.191 after alternative work. Even the most effect-favorable collected sample combination for each pair averaged only +0.606.

The miss was concentrated in one situation

Situation	Forecast	Observed	Error
After baseline-majority work	**0.874**	**0.969**	**-0.094**
After alternative work	**0.584**	**0.078**	**+0.506**

The system slightly underpredicted retention after baseline-majority work. It overpredicted retention by 0.506 after alternative work, where recent work almost completely overturned the baseline majority.

Simple benchmarks

A fixed full-repeat forecast had 16.1 times lower squared error. A pre-existing +0.90 empirical benchmark had 30.3 times lower error and beat the system forecast on 8 of 8 pairs. It was frozen before the confirmation outcomes, but its source overlaps this task set. It is a prospective outside view here, not independent validation.

Pair ranking remains uncertain

The correlation between forecast and outcome was -0.002. A two-sided permutation check gave p = 1.00. This does not show that the forecasts contain no information. It shows that this study found no reliable ranking signal.

Plain reading. The tested system anticipated repetition, but underestimated how strongly recent work would shape its next choice.

4. Replication and context controls

Local Qwen3-4B replication

The local model predicted a mean shift of +0.141. The observed shift was +1.000 across seven dose-three task pairs. After the preferred task, it chose that task again in 28 of 28 usable cells. After the other task, it retained the original preference in 0 of 27 cells.

The local route used greedy decoding and no agent wrapper. One of 56 planned dose-three cells was unusable. We report this nearly complete subset because several dose-one groups had more missing choices.

Standing inside the situation does not help

The comfortable explanation is that the system was asked about a situation that did not exist yet and could not picture it. We tested that. In 80 further sessions the system actually did the work first, then answered the same question, on the same scale, in the same words, with the counterfactual framing removed. No binding choice was ever made in those sessions.

Question asked	Predicted shift	Distance from reality
Before the work existed	+0.290	-0.600
With the finished work in front of it	**+0.223**	**-0.667**
Quoted log, called its own	+0.345	-0.545
Quoted log, called another system's	+0.322	-0.569
What actually happened	+0.891	--

It made the forecast slightly worse. Looking straight at three completed alternative tasks, the system still put the chance of returning to its earlier choice near 0.7; the measured rate was 0.078. All eight pairs underestimated in every condition. The system is not failing to imagine an absent context. It has the evidence in hand and misreads what that evidence will do to it.

Self and observer framings landed 0.024 apart. That matters more here than in the binary control, because here there was room for a gap: both sat near +0.33 against a truth of +0.891, so either could have been far better. Neither was.

Two of three predictions written into the protocol before this run were wrong -- the situated forecast was expected to beat the prospective one and to land nearer the truth. It did neither. Treatment work was fully correct in 92.5% of cells, below the 95% target, as in the main run.

How the work is represented matters

System	Full transcript	Short summary	No transcript
GPT-5.6 Luna in Codex	**+0.725**	**-0.450**	**-0.025**
Qwen3-4B, local	**+1.000**	**+0.950**	**+0.000**

Separate runs changed what remained visible at the next choice. Both systems repeated strongly with the full transcript and did nothing with no transcript. Between those they came apart: telling Luna it had just done a task three times pushed it *away* from that task, reversing the sign of the effect that doing the same work produces. Qwen barely distinguished the two.

For anyone building a way to elicit a model's preferences, that is the practical finding here. The same fact, described rather than shown, can produce the opposite behaviour in the same system. A method that summarises context instead of presenting it is not measuring a weaker version of the same thing.

The Luna run is 240 cells and 1,200 calls. The Qwen run does not record which model produced it, so the comparison between systems rests partly on an artifact with a gap in its record; the Luna reversal does not depend on it. The no-transcript condition is a visible-context control, not a claim that a stateless endpoint erased hidden memory.

Supporting retrospective control

When a model could see the record of recent work, its next-choice prediction was 95.0% accurate. An observer prompt scored 97.5%. A fixed repeat baseline also scored 97.5%. A visible record therefore made next-choice prediction easy. Estimating the size of the effect before the work was harder.

This control saved cell outcomes but not the full provider settings or raw replies now recorded by the primary run. It is supporting evidence.

5. Dependence and checks

Shared-family sensitivity

Task pairs reuse families. Leaving out one family at a time gave mean forecast errors from -0.640 to -0.532. The largest family-disjoint subset contains 4 pairs and has mean error -0.611.

These are descriptive sensitivities. Every original pair already had a negative error. They show that no single family creates the mean result; they are not independent replications.

Checks and provenance

All three treatment tasks were correct in 93.8% of cells. Post-choice task accuracy was 99.2%. Restricting the analysis to the 120 fully correct treatment cells changes the observed shift only to +0.896. Every label and order block had a positive shift.

Seven of eight frozen diagnostic checks passed. The missed threshold required at least 95% of treatment cells to have all three tasks correct; the result was 93.75%. This run is a frozen diagnostic, not a public preregistration.

The runner saved source and protocol hashes before the first model call. They still match. Saved forecast samples, seeds, raw replies, cells, and metrics passed the offline verifier.

6. Relation to other work

Binder et al. (ICLR 2025) trained behavioral self-predictors. This study instead asks for a cold prospective forecast.

Camassa and Shiller (2026) set a user instruction against supplied assistant turns showing a competing pattern, and asked models whether they would hold the instruction. Models scored 83.5% and "systematically underestimate their own resistance to induction pressure": they expected to be swayed more than they were. The systems here miss in the opposite direction, expecting to be swayed less than they were. Their models have an explicit instruction to defend and a supplied history; ours have only an earlier choice and work they actually did. Neither explanation was tested, and the measures differ, so this is a contrast to explain rather than a contradiction.

Qin et al. (ACL 2026) test adaptation without explicit retrieval prompts. Ge et al. (ACL 2026) compare described gambles with passively shown payoff histories. Neither tests this prospective causal forecast.

Singh, Linzen, and Ravfogel (2026) motivate the narrow interpretation: behavioral evidence alone does not establish strong introspection.

7. Limits and reproducibility

Two instruction-tuned systems were tested. The tasks were small and deterministic. Most effects were near the maximum. Task pairs shared families, so pair-level observations were dependent. Codex sampling was not seeded, although each forecast prompt was repeated in five fresh sessions. Only eight pairs entered the confirmation. Supporting controls have less complete provenance. The study does not support claims about all models or subjective experience.

The primary artifact contains the protocol, raw replies, cell-level choices, analysis, model settings, and validation checks. Run `pytest -q`, `validate_research_os_frontier.py`, `winner_protocol/preflight.py`, and `scripts/verify_ranking.py` before any model run.

Repository: github.com/SkyeNygaard/digital-minds

Conclusion. In both tested systems, recent work strongly changed the next binding choice. Their forecasts anticipated some repetition but underestimated its strength.